Bavish Kulur

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Research Interests: Deep Reinforcement Learning, Long Horizon RL, Post-Training, Representation Learning, Skill Learning

Education

University of Alberta Sept 2025 – May 2027 (Expected)
MSc (Thesis), Computing Science **GPA: 4.0/4.0**
Advisor: Prof. Marlos C. Machado
Coursework: Reinforcement Learning I & II; Foundations of Reasoning in LLMs; Machine Learning II

Indian Institute of Technology Bombay 2017 – 2022
Dual Degree: BTech, Mechanical Engineering; MTech, AI & Data Science **GPA: 8.84/10**
Coursework: Advanced Machine Learning; Foundations of Intelligent and Learning Agents; Learning over Graphs
Minor in Computer Science and Engineering

Publication

PREFINE: Preference-Based Implicit Reward and Cost Fine-Tuning for Safety Alignment
Richa Verma, **Bavish Kulur**, Sanjay Chawla, Balaraman Ravindran
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2026. Paper Code

Research Projects

Online Skill Discovery with Laplacian Representations Sept 2025 – Present
University of Alberta | Advisor: Prof. Marlos C. Machado

- Investigating how Laplacian representations can drive diverse skill learning without task rewards, using representation changes as rewards for skill-conditioned policies.
- Comparing Laplacian-based skills with METRA in continuous-control environments; studying how eigenvalue scaling and skill sampling affect spatial coverage and directional balance.
- Analyzing why representations that capture global spatial structure can yield weak local learning signals; evaluating multi-step SAC returns to improve credit assignment over longer horizons.

PREFINE: Safe RL through Direct Preference Optimization 2024 – 2025
Advisors: Prof. Balaraman Ravindran (IIT Madras) and Dr. Sanjay Chawla (QCRI, Qatar)

- Coauthored a method adapting Direct Preference Optimization (DPO) to trajectory-level preferences, fine-tuning pretrained continuous-control policies for safety while retaining reward performance.
- The method uses policy-sampled counterfactual actions for preference comparisons; the study reports over 60% fewer constraint violations and catastrophic failures compared to baselines. Published at AAMAS 2026.

GARE: Greedy Actor Regularized Expert Sept – Dec 2025
University of Alberta | Course project with Prof. Martha White | Report Code

- Proposed a value-based RL algorithm for continuous actions that initializes action search with a learned proposal function and refines it through gradient ascent to construct TD targets.
- Evaluated against GreedyAC and Soft Actor-Critic (SAC), outperforming both baselines in the project experiments.

Fairness Evaluation of Distilled Language Models 2024 – 2025
Advisors: Prof. Balaraman Ravindran (IIT Madras) and Prof. Karthik Sheshadri (NIT AP)

- Investigated how knowledge distillation from large language models affects fairness in task-specific student models, extending evaluation beyond predictive accuracy and computational cost.

Microstructure Segmentation in CT Images 2021 – 2022

IIT Bombay | Thesis project with Prof. Shyamprasad Karagadde

- Trained U-Net models to segment aluminum-alloy microstructures in real-world CT images and quantify material properties, including porosity.
- Integrated the trained model into ImageJ to support segmentation and quantitative analysis within an established scientific imaging tool.

GPU and CPU Parallelization for Stochastic Finite Element Analysis Jan – Apr 2021

IIT Bombay | Course project with Prof. S. Gopalakrishnan

- Parallelized the response surface algorithm using custom CUDA kernels and OpenMP multithreading, **reducing computation time by 35%**.

Professional Experience

Post-Baccalaureate Research Fellow, IIT Madras Nov 2023 – Jun 2025

Robert Bosch Centre for Data Science and AI | Advisor: Prof. Balaraman Ravindran

Product Engineer (SDE 1), Sprinklr Gurugram, India Jul 2022 – Feb 2023

Software Engineering Intern, Arturo Technologies Bangalore, India Mar – Jun 2019

Technical Skills

Machine learning: Python, JAX, PyTorch

Systems and parallel computing: C++, CUDA, OpenMP, Linux, Docker, Git, Slurm

Teaching Assistantships

Reinforcement Learning (CMPUT 356), University of Alberta	Fall 2025, 2026
File and Database Management (CMPUT 291), University of Alberta	Winter 2026
Introduction to Machine Learning, NPTEL MOOC	Jan – Apr 2025
Computer Programming and Utilization (CS 101), IIT Bombay	Spring 2019

Selected Honors

Selected for the Reinforcement Learning Workshop, Indian Institute of Science	2024
Student Innovation Grant, ARTPARK, Indian Institute of Science	2021
Selected for a semester exchange at the National University of Singapore	2020
Kishore Vaigyanik Protsahan Yojana (KVPY) award, Government of India	2017
Qualified in the Regional Mathematical Olympiad and attended the Indian National Mathematics Olympiad training camp at Indian Statistical Institute Bangalore	2015

Activities

Convener, Web and Coding Club, IIT Bombay 2018 – 2019

Developer and Founding Member, InstiApp, Developers Community, IIT Bombay 2018 – 2019